
EE 306: Computer Organization & Microprocessor

Course Code	EE 306
Course Name	Computer Organization & Microprocessor
L-T-P-C	3-0-0-3
One-time approval	5th Senate (Oct 2013)

Course Contents

- **Digital Logic and Digital Systems:** Overview and history of computer architecture, combinational vs sequential logic, hardware description languages (VHDL), physical constraints (gate delay, fan-in, fan-out, energy/power).
- **Instruction Set Architecture:** Introduction to instruction set architecture, Basic organization of computing machine: fetch, decode, and execute; Instruction set types, instruction format, addressing modes, subroutine call and return mechanisms; Structure of machine-level programs; Low-level architectural support for high level languages. Performance assessment.
- **Computer Arithmetic:** Representation of numeric data, signed and unsigned arithmetic; Range, precision and errors in floating-point arithmetic; Design of arithmetic and logic unit (ALU).
- **Processor Architecture:** CISC vs RISC Designs, simple implementation schemes, datapath design, control unit: hardwired realization vs micro-programmed realization, multi-cycle implementation. Instruction level parallelism, instruction pipelining, pipeline hazards.
- **Memory Architecture:** Storage systems, introduction to memory hierarchy: importance of temporal and spatial locality; main memory organization, cache memory: address mapping, block size, replacement, and store policies; virtual memory system: page table and TLB.
- **Interfacing and I/O Organization:** External storage; IO fundamentals: handshaking, buffering, programmed IO, interrupt driven IO; Interrupt handling mechanism, Buses: protocols, arbitration, direct memory access (DMA).

Text Books

1. DA Patterson and JL Hennessy, **Computer Organization and Design**, 4th Edition, Morgan Kaufmann Publisher, 2010.

References

1. J.P. Hayes, **Computer Architecture and Organization**, Mc Graw Hill.
2. A.S. Tanenbaum, **Structured Computer Organization**, PHI Publication.
3. W. Stalling, **Computer Organization and Architecture**, PHI Publication.